

Companion EQS — FollowSlot

High-level overview · AI-Companion-Prototype branch

Status	WORKING	Asset	EQS_FollowSlot.uasset
--------	---------	-------	-----------------------

What it does

Picks where the companion should stand relative to the player while following. Every half-second, the BT task asks the EQS "given the player, what's the best slot to occupy right now?" and the EQS returns one location. The companion walks to that slot instead of a hardcoded rear-right formation offset.

How it runs

- Assigned on the *Follow Player* BT node, under the **EQS** category.
- Queries every 0.5s (designer-tunable), throttled so only one is in flight at a time.
- Only fires while the companion is actively moving toward the player — idles when close.
- Receives the player actor as its anchor via `EnvQueryContext_Player`.
- Runs in **SingleResult** mode — returns one winning location, not a ranked list.

How it decides the best slot

The query asset (designer-authored) runs three stages on candidate points around the player:

- **Generator** — spits out candidate positions (typically a donut around the player at formation radius).
- **Filters** — hard reject points: must be on the navmesh, must not overlap actors, etc.
- **Scorers** — soft weights for distance-to-player, line-of-sight, distance-from-self (anti-oscillation), and optionally 'behind player's movement direction'.

Each surviving candidate gets a total score from the weighted scorers. The highest score wins. Tuning the weights changes personality: heavy LOS = always visible to player; heavy distance-from-self = sticky slot, no ping-ponging; heavy behind-velocity = covers the player's six.

Fallback (why it can't fully break)

If the query asset is unset, in-flight, or returns no items, the task falls back to C++ formation math — rear-right offset of the player's velocity vector. So a mistuned or missing EQS degrades into "decent fixed formation" rather than broken follow behaviour.

Tuning surface

- **C++ side** — `FollowSlotQuery` (which asset) and `EqsQueryInterval` (how often) on the BT node.
- **Asset side** — generator radius, filter strictness, scorer weights, all editable in the EQS editor.
- **Anchor side** — `EnvQueryContext_Player` reads the player from the blackboard; no tuning needed.